

Indraprastha Institute of Information Technology

Prior-Guided KernelSHAP for Explainable Image Classification

Classic KernelSHAP vs Grad-CAM++ Prior-Guided
Importance-Corrected WLS KernelSHAP

B.Tech Project Report

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Submitted by:

Madhav Gaur

Roll No.: 2023303

Under the guidance of:

Anubha Gupta

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Declaration

I, **Madhav Gaur** (Roll No. **2023303**), hereby declare that the work presented in this report titled “**Prior-Guided KernelSHAP for Explainable Image Classification**” is an original work carried out by me under the supervision of **Anubha Gupta**. This work has not been submitted, in part or in full, for the award of any degree or diploma at any institute or university.

Place: New Delhi

Date: 20/04/2026

Signature: _____

Name: Madhav Gaur

Certificate

This is to certify that the project report entitled “**Prior-Guided KernelSHAP for Explainable Image Classification**” submitted by **Madhav Gaur** (Roll No. **2023303**) is a bonafide record of work carried out under my supervision and guidance, and is correct to the best of my knowledge.

Anubha Gupta

Indraprastha Institute of Information Technology

Place: New Delhi

Date: 20/04/2026

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Abstract

KernelSHAP provides a model-agnostic approximation of Shapley values via a weighted linear regression over randomly sampled feature coalitions. However, for vision tasks the feature dimension can be large, making naive pixel-level KernelSHAP computationally expensive. This report implements and evaluates two superpixel-based explainability pipelines: (i) **Classic KernelSHAP** using `shap.KernelExplainer` over SLIC superpixel features, and (ii) a **Grad-CAM++ prior-guided, importance-corrected WLS KernelSHAP** sampler that biases coalition generation toward superpixels ranked salient by Grad-CAM++ while correcting for proposal bias in the regression weights. A key theoretical contribution is a formal proof that the importance-corrected estimator is *unbiased* with respect to the KernelSHAP objective: the proposal distribution $q(z)$ cancels exactly via a change-of-measure argument, and the prior acts purely as a variance-reduction device. Experiments are conducted on **ImageNet** validation images using a pretrained ResNet-18. Results show substantial runtime improvements for the prior-guided sampler while preserving similarity to classic KernelSHAP, measured using Spearman correlation at superpixel and pixel-density levels. The project also standardizes artifact generation (original image, Grad-CAM++ map, classic SHAP heatmap, prior-guided SHAP heatmap) and logs timings and metrics for reproducibility.

Code Repository: [GITHUB](#)

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Chapter 1

Introduction

1.1 Motivation

Deep learning models for image classification often behave as black boxes: they can achieve high accuracy, yet provide little insight into *why* a particular class is predicted. Explainability methods address this gap by attributing the prediction to regions of the input image, which helps practitioners interpret model decisions, debug spurious correlations, detect dataset bias, and build user trust.

Among popular attribution methods, **SHAP** (Shapley Additive Explanations) is especially desirable because its feature attributions are grounded in Shapley values from cooperative game theory and satisfy strong axiomatic properties: *(i) efficiency* (attributions add up to the model output difference), *(ii) symmetry* (equivalent features receive equal credit), *(iii) dummy* (irrelevant features get zero attribution), and *(iv) additivity* (explanations are consistent under model addition). These properties make SHAP explanations more principled and comparable across inputs and models than many faster alternatives such as **Grad-CAM++**, **LIME**, or **DeepLIFT**, which can be more heuristic, sensitive to implementation choices, or less directly tied to a well-defined attribution objective.

However, the most general and “vanilla” SHAP estimator, **KernelSHAP**, is *computationally expensive* for images. KernelSHAP requires evaluating the model on many randomly sampled coalitions of features; when features are pixels (or even superpixels), the coalition space grows combinatorially and the number of model evaluations needed for a stable estimate becomes large. This makes naive image-level KernelSHAP slow in practice, motivating approaches such as using superpixels and/or leveraging informative priors to guide coalition sampling.

There also exist SHAP variants tailored to specific model classes (e.g., **DeepSHAP** for neural networks), which can be substantially faster than KernelSHAP by exploiting network structure and backpropagation-style computations. Nevertheless, DeepSHAP

is less commonly used as a default for image explainability in many practical pipelines compared to saliency approaches (e.g., Grad-CAM-family methods) or superpixel-based model-agnostic methods, partly due to differences in assumptions, implementation complexity, and the desire for fully model-agnostic explanations.

1.2 Problem Statement

Classic KernelSHAP becomes expensive when applied directly to pixels due to the large number of features. A standard workaround is to use **superpixels** as features, but sampling and regression can still be slow. This project explores a **prior-guided coalition sampler** using Grad-CAM++ as a saliency prior to accelerate KernelSHAP while keeping the estimator faithful through importance correction.

1.3 Contributions

- Implemented **classic KernelSHAP** with SLIC superpixel features using `shap.KernelExplainer`.
- Implemented a **Grad-CAM++ prior-guided** coalition proposal with **importance-corrected** kernel weights, and formally proved that the resulting estimator is unbiased with respect to the KernelSHAP objective (Proposition 4.1).
- Built an end-to-end evaluation pipeline for **ImageNet** images with artifact saving, timing logs, and Spearman correlation.
- Conducted a **convergence study** (405 KernelSHAP calls across 9 sample budgets $\times$ 9 images $\times$ 5 seeds) with Insertion/Deletion AUC and run-to-run $\bar{\sigma}$ diagnostics to select a stable reference budget $N^* = 12,000$.

Chapter 2

Background

2.1 Shapley Values and Additivity

Given a set of features $N = \{1, \dots, M\}$ and a value function $f(S)$ (model output under coalition S), the Shapley value for feature i is:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (M - |S| - 1)!}{M!} (f(S \cup \{i\}) - f(S)).$$

These values are uniquely defined by the four axioms of efficiency, symmetry, dummy, and additivity, and represent the only fair attribution of the model output to individual features. However, computing them exactly requires summing over all 2^M coalitions, which is infeasible for even moderate M .

2.2 KernelSHAP: An Estimator, Not Exact Shapley Values

It is important to note that **KernelSHAP does not compute true Shapley values**. Rather, it is a *Monte Carlo estimator* that approximates Shapley values by solving a weighted linear regression over a finite sample of coalitions. The quality of the approximation depends on the number of sampled coalitions (`nsamples`): with more samples the estimates converge toward the true Shapley values, but convergence can be slow for large M .

This estimation nature has two important practical consequences:

- **Variance:** Two runs of KernelSHAP on the same input with the same budget will generally produce slightly different attribution vectors due to Monte Carlo noise in the sampled coalitions.

- **Bias:** With limited `nsamples`, the weighted regression is fit on an incomplete picture of the coalition space, introducing systematic bias. The SHAP kernel weights are designed to reduce this bias by emphasizing small and large coalitions, but they cannot eliminate it at finite sample sizes.

Because KernelSHAP is an estimator, any modification to the coalition sampling procedure (such as the prior-guided approach developed in this work) must be evaluated relative to the KernelSHAP estimate rather than true Shapley values. The relevant question is not whether the prior-guided method recovers exact Shapley values, but whether it produces attributions that are *similar* to those of classic KernelSHAP at a lower computational cost.

2.2.1 Coalition Sampling and `nsamples`

The exact Shapley values require summing over all 2^M coalitions, which is infeasible even for moderate M . KernelSHAP therefore draws a finite set of coalitions

$$\{z^{(j)}\}_{j=1}^N, \quad N = \text{nsamples},$$

and evaluates the model outputs $\{f(z^{(j)})\}$ to form the regression dataset.

In practice, coalition sizes are sampled more often near the extremes (small s and large s) because the SHAP kernel puts higher mass there. A common size distribution induced by the kernel is

$$\Pr(s) \propto \frac{1}{s(M-s)}, \quad s \in \{1, \dots, M-1\},$$

followed by sampling a subset of size s uniformly among all $\binom{M}{s}$ subsets.

2.2.2 SHAP Kernel Weights

Let $s = \sum_{i=1}^M z_i$ be the coalition size. KernelSHAP assigns a kernel weight

$$w(z) \propto \frac{M-1}{\binom{M}{s} s(M-s)} \quad \text{for } 1 \leq s \leq M-1,$$

and typically enforces the two anchor points $z = \mathbf{0}$ and $z = \mathbf{1}$ with very large weight to stabilize the fit and enforce additivity.

2.2.3 Paired Complements

To reduce variance, we can pair a sampled coalition z with its complement $(\mathbf{1} - z)$. This increases coverage of both small and large coalitions while preserving the kernel structure.

2.2.4 WLS Solution

Let $X \in \mathbb{R}^{(N+2) \times (M+1)}$ be the design matrix formed by concatenating a bias column of ones and the coalition vectors (including anchors $z = \mathbf{0}$ and $z = \mathbf{1}$), and let $y \in \mathbb{R}^{N+2}$ be the corresponding outputs. The WLS estimate is obtained by

$$\hat{\beta} = \arg \min_{\beta} \sum_j w(z^{(j)}) (y^{(j)} - X^{(j)} \beta)^2,$$

where $\beta = (\phi_0, \phi_1, \dots, \phi_M)^\top$.

2.3 SLIC Superpixels as Features

For images, treating each pixel as a feature is computationally expensive and unstable. We therefore compress the input into M interpretable regions using SLIC (Simple Linear Iterative Clustering), and treat each *superpixel* as a binary feature.

2.3.1 Terminology

SLIC outputs a label map

$$\mathbf{segments} \in \{0, 1, \dots, M-1\}^{H \times W},$$

where $\mathbf{segments}(u, v) = k$ assigns pixel (u, v) to superpixel k . Each superpixel corresponds to the set

$$\Omega_k = \{(u, v) : \mathbf{segments}(u, v) = k\}.$$

2.3.2 SLIC Objective and Parameters

SLIC performs a k -means-like clustering in a joint color-spatial space. In each iteration, pixels are assigned to the nearest cluster center based on a distance of the form

$$D = \sqrt{d_c^2 + \left(\frac{m}{S}\right)^2 d_s^2},$$

where:

- d_c is the color distance (e.g., in RGB or Lab space),
- d_s is the spatial distance in the image plane,
- $S \approx \sqrt{\frac{HW}{M}}$ is the grid interval between initial centers,

- m is the **compactness** parameter controlling the trade-off between color similarity and spatial regularity.

The algorithm alternates assignment and center update until convergence, producing approximately M spatially contiguous segments that respect boundaries.

2.3.3 Superpixel Masking Game

We define a binary coalition vector $z \in \{0, 1\}^M$ over superpixels. A masked image is created by replacing pixels in OFF superpixels ($z_k = 0$) with baseline pixels, while keeping pixels in ON superpixels ($z_k = 1$) unchanged:

$$x^{(z)}(u, v) = \begin{cases} x(u, v), & \text{if } z_{\text{segments}(u, v)} = 1, \\ x_{\text{base}}(u, v), & \text{if } z_{\text{segments}(u, v)} = 0. \end{cases}$$

The value function is then $f(z) = \text{model}(x^{(z)})$ for the target logit (or probability) being explained.

2.4 Motivation for Prior-Guided Sampling

Uniform coalition sampling in KernelSHAP wastes model evaluations on uninformative masks: a coalition drawn uniformly at random from $\{0, 1\}^M$ has no preference for including superpixels that actually influence the prediction. A saliency prior — here provided by Grad-CAM++ — tells us, cheaply and without additional model queries beyond one backward pass, which spatial regions are likely responsible for the top-1 class score. By oversampling coalitions that include these high-saliency superpixels, we obtain a more informative regression dataset per unit compute.

Crucially, oversampling alone would bias the WLS regression away from the KernelSHAP objective. The *importance correction* (dividing the SHAP kernel weight by the proposal probability $q(z)$) removes this bias exactly via a change-of-measure argument, so the fitted regression still targets the same weighted least-squares problem as classic KernelSHAP. This is made precise in Proposition 4.1 in Chapter 4. The prior therefore acts as a *variance-reduction device* rather than a change in objective.

Chapter 3

Methodology

3.1 Dataset and Model

3.1.1 ImageNet

We evaluate on a representative subset of **ImageNet** ILSVRC 2012 validation images (224×224 RGB). These images span a diverse range of object categories and scene complexities, providing a challenging and well-established benchmark for assessing both explanation quality and runtime.

3.1.2 Model

We use a **ResNet-18 pre-trained on ImageNet** (via `torchvision.models`) and explain the **top-1 predicted class** logit in all experiments. The model is used as-is without any fine-tuning; images are resized to 224×224 and normalised with ImageNet mean/std before being passed to the network.

3.2 Masking Baseline and Feature Space

3.2.1 Normalized Image Space

Let x be the input image in normalized space (ImageNet mean/std). The baseline x_0 is a Gaussian-blurred version of the input image (blur radius = 11 px), also normalised with ImageNet mean/std. This produces a more naturalistic counterfactual than a zero baseline: OFF superpixels appear blurred rather than replaced by a uniform colour, reducing out-of-distribution artefacts during model evaluation. A coalition mask z is lifted to pixel space using the superpixel-to-pixel membership matrix.

3.2.2 Superpixel Features

We segment the image into M superpixels using SLIC. Each coalition $z \in \{0, 1\}^M$ defines a binary spatial mask $m(z) \in \{0, 1\}^{H \times W}$:

$$x(z) = x_0 + m(z) \odot (x - x_0).$$

3.3 Establishing a Reference Baseline with Classic KernelSHAP

As discussed in Section 2.2, KernelSHAP is itself a Monte Carlo estimator and does not return exact Shapley values. This means any comparison between our prior-guided method and a “reference” explanation must be made carefully: we cannot compare against ground-truth Shapley values (intractable), so we instead compare against **classic KernelSHAP run at a high sample budget**, treating it as a stable empirical reference.

The key question is therefore: *how do we obtain a reliable baseline estimate?*

3.3.1 Finding a Good Baseline

Because KernelSHAP is a Monte Carlo estimator, the attribution vector $\phi^{(N)} \in \mathbb{R}^M$ depends on both the coalition budget $N = \text{nsamples}$ and the random seed used during sampling. We therefore need to answer two concrete questions before fixing a reference budget:

1. *Does the explanation stabilise?* As N grows, does $\phi^{(N)}$ converge to a consistent vector, and at what budget?
2. *Does the explanation remain meaningful?* Does the resulting heatmap genuinely localise the features the model uses?

To answer these, we run a dedicated convergence study over 9 ImageNet images.

Sample-Budget Sweep

We sweep over nine values of `nsamples`:

$$N \in \{100, 500, 1000, 2000, 4000, 6000, 8000, 10000, 12000\}.$$

At each value of N , KernelSHAP is run $N_{\text{runs}} = 5$ independent times per image (different random seeds), yielding an attribution vector $\phi^{(N,r)} \in \mathbb{R}^M$ and a wall-clock time per run r . The study therefore executes $9 \times 9 \times 5 = 405$ KernelSHAP calls in total.

Convergence Diagnostic 1 — Insertion / Deletion AUC

To assess meaningfulness, each ϕ is evaluated using the **insertion** and **deletion** protocols. The pixel-level attribution map is first obtained via the sum-preserving superpixel-to-pixel expansion $A(p) = \phi_k / |\Omega_k|$ for $p \in \Omega_k$.

- **Deletion.** Starting from the original image, pixels are replaced by the blur baseline in decreasing order of $A(p)$. The top-1 logit is recorded at $N_{\text{steps}} = 60$ evenly-spaced removal fractions. A lower Deletion AUC indicates a more faithful explanation.
- **Insertion.** Starting from the blur baseline, pixels are revealed in the same order. A higher Insertion AUC indicates a more faithful explanation.

Both AUCs are computed via the trapezoidal rule and logged for every (image, N , run) entry.

Convergence Diagnostic 2 — Run-to-Run SHAP Vector Variance

To assess stability, for each image and each N we stack the five attribution vectors into $\Phi \in \mathbb{R}^{5 \times M}$ and compute the **mean per-superpixel standard deviation** across runs:

$$\bar{\sigma}(N) = \frac{1}{M} \sum_{k=1}^M \text{std}(\Phi_{:,k}).$$

A plot of $\bar{\sigma}(N)$ versus N on a log scale reveals the elbow beyond which increasing the budget yields negligible additional stability.

Additivity Check

Each run also records the efficiency error $\Delta_{\text{eff}} = |\sum_k \phi_k - (f(\mathbf{1}) - f(\mathbf{0}))|$, which should remain close to zero at all budgets since KernelSHAP enforces additivity via its anchor points. A large Δ_{eff} at a given N flags numerical instability in the weighted regression.

Baseline Budget Selection

The reference budget N^* is chosen conservatively as the largest value in the sweep, $N^* = 12,000$, to ensure the classic KernelSHAP estimates are as close as possible to the true Shapley values and serve as a reliable reference for all subsequent comparisons.

3.4 Classic KernelSHAP on Superpixels

We use `shap.KernelExplainer` with feature vectors:

$$z = \mathbf{0} \quad (\text{background}), \quad z = \mathbf{1} \quad (\text{instance}).$$

The explainer samples additional z internally and solves the weighted regression.

3.5 Grad-CAM++ and the Prior-Guided Coalition Sampler

3.5.1 How Grad-CAM++ Works

Consider a CNN with last convolutional feature maps $\{A^k\}$ and class score y^c for class c (typically the logit). Grad-CAM++ computes a weighted combination of activations:

$$L^c(u, v) = \text{ReLU} \left(\sum_k w_k^c A^k(u, v) \right),$$

where the weights w_k^c are derived from gradients of y^c with respect to A^k . Unlike Grad-CAM, Grad-CAM++ uses pixel-wise weights $\alpha_k^c(u, v)$ based on higher-order gradient terms, which improves localization when multiple instances of the same class are present. The result is a heatmap $L^c \in \mathbb{R}^{H \times W}$ that can be normalized to $[0, 1]$ for visualization.

3.5.2 From Grad-CAM++ to a Superpixel Prior

To translate the continuous heatmap into a prior over superpixels, we score each superpixel by the mean CAM value:

$$\text{score}(k) = \frac{1}{|\Omega_k|} \sum_{(u,v) \in \Omega_k} L^c(u, v).$$

We then rank superpixels by $\text{score}(k)$ and split them into two disjoint sets:

- **Prior pool $\mathcal{P}$:** top ρ fraction of superpixels (high CAM support),
- **Rest pool $\mathcal{R}$:** remaining superpixels.

Here $\rho \in (0, 1)$ is a tunable hyperparameter (e.g., `prior_pool_frac`).

3.5.3 Prior-Biased Coalition Sampling with Importance Correction

We still aim to estimate the *KernelSHAP* solution, i.e., the regression under the SHAP kernel. However, to reduce the number of expensive model evaluations, we sample coalitions from a biased proposal distribution $q(z)$ that prefers including superpixels from $\mathcal{P}$:

1. Sample a coalition size s using the SHAP-induced size mass:

$$\Pr(s) \propto \frac{1}{s(M-s)}, \quad s \in \{1, \dots, M-1\}.$$

2. Choose how many ON features come from the prior pool: sample k in the feasible range $k \in [\max(0, s - |\mathcal{R}|), \min(s, |\mathcal{P}|)]$ using a (truncated) binomial-like rule controlled by a parameter p (e.g., `prior_pick_frac`).
3. Sample k superpixels uniformly from $\mathcal{P}$ and $(s - k)$ uniformly from $\mathcal{R}$ to form z .

Since coalitions are no longer sampled exactly according to the SHAP kernel, we apply importance correction in the WLS weights:

$$w^*(z) \propto \frac{w_{\text{SHAP}}(z)}{q(z)}.$$

This keeps the regression objective aligned with KernelSHAP while allowing aggressive sampling toward high-prior coalitions. In addition, we optionally pair each sampled coalition with its complement to improve coverage and reduce variance.

3.5.4 Algorithm

Algorithm 1 gives a complete pseudocode description of the prior-guided KernelSHAP procedure. The algorithm has three distinct phases: a *setup phase* (Steps 1–7) that is cheap and requires only one forward–backward pass; a *sampling phase* (Steps 8–15) that builds the coalition set and records log-probabilities; and an *inference phase* (Steps 16–22) that evaluates the model in batches and solves a constrained WLS regression with numerically stable, log-space weight normalisation.

Several design choices in the algorithm deserve attention:

- **No explicit anchor rows.** Rather than inserting $z = \mathbf{0}$ and $z = \mathbf{1}$ as high-weight rows in the regression matrix (as `shap.KernelExplainer` does internally), the code evaluates them once to obtain $f_{\emptyset} = f(x^{(0)})$ and $f_{\text{full}} = f(x^{(1)})$ as scalar constants. Additivity is then enforced *directly* via a Lagrange-multiplier constraint in the WLS solve (Step 22), guaranteeing $\sum_k \phi_k = f_{\text{full}} - f_{\emptyset}$ exactly.
- **Two-phase loop.** Sampling and model evaluation are deliberately separated: all N coalition vectors and their log-probabilities are collected first (Steps 8–15), and the model is then queried in a single batched pass (Step 16). This maximises GPU utilisation.
- **Log-space weight normalisation.** Corrected weights are computed in log-space as $\log w_i^* = \log \pi(z_i) - \log q(z_i)$ and then normalised by subtracting the maximum

before exponentiating (Step 18). This prevents numerical overflow when kernel weights span many orders of magnitude.

Algorithm 1 Prior-Guided Importance-Corrected Superpixel KernelSHAP

Require: Image x , classifier f , baseline $x^{(0)}$, superpixel count M , prior budget N_p , pool fraction ρ_{pool} , pick fraction ρ_{pick} , complement flag, batch size B

Ensure: SHAP values $\phi \in \mathbb{R}^M$, pixel attribution map A

- 1: Segment x into SLIC superpixels $S_1, \dots, S_M$
- 2: Compute target class $c \leftarrow \arg \max_k f_k(x)$ and Grad-CAM++ prior map G
- 3: **for** $j = 1, \dots, M$ **do**
- 4: $g_j \leftarrow \frac{1}{|S_j|} \sum_{p \in S_j} G(p)$
- 5: **end for**
- 6: Rank superpixels by g_j in descending order
- 7: $P \leftarrow \text{round}(\rho_{\text{pool}} \cdot M)$; define **priorPool** (top- P) and **restPool** (remainder)
- 8: $f_{\emptyset} \leftarrow f_c(x^{(0)})$, $f_{\text{full}} \leftarrow f_c(x)$
- 9: Define $q_s(s) \propto \frac{1}{s(M-s)}$ for $s = 1, \dots, M-1$
- 10: Initialise coalition list Z , log-probability list L
- 11: **for** $t = 1, \dots, N_p$ **do**
- 12: Sample coalition size $s \sim q_s$
- 13: Sample $k \in \{\max(0, s - P), \dots, \min(s, P)\}$ using ρ_{pick}
- 14: Sample k superpixels from **priorPool** and $s - k$ from **restPool**; form $z \in \{0, 1\}^M$
- 15: Compute $\log q(z)$ analytically and append $(z, \log q(z))$ to (Z, L)
- 16: **if** complement flag **and** $0 < M - s < M$ **then**
- 17: Append complement coalition $\mathbf{1} - z$ with its $\log q$ to (Z, L)
- 18: **end if**
- 19: **end for**
- 20: Evaluate model in batches of size B : $y_i \leftarrow f_c(x(z_i))$ for each $z_i \in Z$
- 21: For each i : $\log w_i \leftarrow \log \pi(z_i) - \log q(z_i)$ where $\pi(z) = \frac{\binom{M-1}{s}}{\binom{M}{s} s(M-s)}$
- 22: Normalise: $w_i \leftarrow \exp(\log w_i - \max_j \log w_j)$
- 23: Solve constrained WLS:

$$\phi \leftarrow \arg \min_{\varphi} \sum_i w_i (y_i - f_{\emptyset} - z_i^\top \varphi)^2 \quad \text{s.t.} \quad \sum_{j=1}^M \varphi_j = f_{\text{full}} - f_{\emptyset}$$

- 24: Construct pixel map: $A(p) = \phi_j / |S_j|$ for $p \in S_j$
 - 25: **return** ϕ, A
-

3.6 Evaluation Metrics

- **Runtime:** Classic time vs prior-guided time (wall-clock seconds).
- **Additivity check:** Compare $f(\mathbf{1}) - f(\mathbf{0})$ with $\sum_i \phi_i$; the gap Δ_{eff} should be close to zero.
- **Spearman correlation:** Between ϕ^{classic} and ϕ^{prior} at the superpixel level.
- **MSE:** Mean squared error $\frac{1}{M} \|\phi^{\text{classic}} - \phi^{\text{prior}}\|^2$ between the two attribution vectors.
- **L2 distance:** $\|\phi^{\text{classic}} - \phi^{\text{prior}}\|_2$ at the superpixel level and $\|A^{\text{classic}} - A^{\text{prior}}\|_2$ at the pixel level, where A is the sum-preserving pixel expansion of ϕ .
- **Insertion / Deletion AUC:** Faithfulness of each method’s heatmap evaluated independently, with a higher Insertion AUC and lower Deletion AUC indicating a more faithful explanation.

Table 3.1 summarises all metrics together with their interpretation direction.

Table 3.1: Summary of evaluation metrics and their interpretation.

Metric	What it measures	Better
Runtime (s)	Wall-clock inference time	↓
Additivity gap Δ	WLS conditioning / sample coverage	≈ 0
Spearman (superpixel)	Rank agreement of ϕ vectors	↑
Spearman (pixel)	Spatial rank agreement of heatmaps	↑
MSE	Squared attribution error	↓
L2 distance	Absolute attribution error	↓
Insertion AUC	Faithfulness (reveal salient first)	↑
Deletion AUC	Faithfulness (remove salient first)	↓

Chapter 4

Implementation Details

4.1 Pipeline Overview

We implement and compare two explanation pipelines that operate on the *same* superpixel feature space. Both pipelines use identical SLIC parameters to ensure compatibility (same segmentation, same feature set), and differ only in how coalitions are sampled and how Shapley values are estimated.

1. Load an image x ; compute the model’s logits and top- K predicted classes.
2. Compute SLIC superpixels; obtain a label map `segments` and construct the superpixel membership matrix.
3. **Method 1 (Classic KernelSHAP)**: run `shap.KernelExplainer` on superpixel masks to estimate SHAP values.
4. Compute Grad-CAM++ for the top-1 class; convert the CAM heatmap into a ranked list of superpixels and define a *prior pool*.
5. **Method 2 (Prior-guided KernelSHAP)**: sample coalitions from a biased proposal $q(z)$, evaluate the model, and solve an importance-corrected WLS regression to match the KernelSHAP objective.
6. Run an outer loop over the image dataset and log per-image artifacts + metrics (timings, additivity checks, and Spearman correlations) into a CSV.

Shared Superpixel Feature Space (Fair Comparison)

To ensure both methods explain the same game, we keep SLIC hyperparameters fixed across both pipelines:

```
n_segments = SLIC_NSEG, compactness = SLIC_COMP, sigma = SLIC_SIG.
```

Thus, both methods operate on the same number of features M (the realized number of superpixels) for a given image.

Masking Baseline (Consistency)

Each coalition vector $z \in \{0, 1\}^M$ defines a masked image $x^{(z)}$ by keeping ON superpixels from the original image and replacing OFF superpixels with baseline pixels. This baseline choice is kept consistent within the experiment to preserve comparability of $f(\mathbf{1}) - f(\mathbf{0})$ and the additivity gap.

4.2 Experimental Hyperparameters

Table 4.1 centralises all fixed hyperparameters used in the main experiment.

Table 4.1: Experimental setup used in the two evaluation cases.

Parameter	Value	Description
SLIC_NSEG	50	Target number of superpixels
SLIC_COMP	10	SLIC compactness
SLIC_SIG	1	SLIC Gaussian smoothing σ
blur_radius	11 px	Masking baseline blur radius
prior_pool_frac	0.30	Fraction of superpixels in prior pool $\mathcal{P}$
prior_pick_frac	0.60	Fraction of ON features drawn from $\mathcal{P}$
batch_size	64	Masked images per forward pass
num_images	72	Total number of images, covering both low- and high-confidence samples

We conduct two experimental runs while keeping all hyperparameters fixed except the coalition budgets. In both cases, the evaluation is performed on 72 images spanning both low-confidence and high-confidence predictions.

- **Case 1 (converged baseline):** `nsamples_classic` = 12,000 and `nsamples_prior` = 3,000. Here, the classic KernelSHAP estimate is treated as a reasonable proxy for ground truth since it lies in the convergence regime.
- **Case 2 (non-converged baseline):** `nsamples_classic` = 300 and `nsamples_prior` = 300. Here, the classic KernelSHAP estimate itself is outside the convergence regime and is therefore a weak proxy for ground truth.

Apart from the change in coalition budgets across these two cases, all other experimental settings remain identical.

4.3 Method 1: Classic KernelSHAP (Superpixel Features)

4.3.1 What We Estimate

Classic KernelSHAP approximates Shapley values by fitting a linear surrogate on coalition samples:

$$f(z) \approx \phi_0 + \sum_{i=1}^M z_i \phi_i,$$

where $z_i = 1$ means superpixel i is ON. The target $f(z)$ is typically the logit (or probability) of the class being explained.

4.3.2 Coalition Sampling (nsamples)

The exact Shapley computation requires 2^M coalitions. KernelSHAP instead draws a finite set

$$\{z^{(j)}\}_{j=1}^N, \quad N = \text{nsamples_classic},$$

evaluates $\{f(z^{(j)})\}$, and uses them in a weighted regression. In practice, KernelSHAP emphasizes small and large coalition sizes (near 0 and M) because those regimes receive higher weight under the SHAP kernel.

4.3.3 SHAP Kernel Weights and Anchors

Let $s = \sum_{i=1}^M z_i$ be the coalition size. KernelSHAP assigns a weight:

$$w_{\text{SHAP}}(z) \propto \frac{M-1}{\binom{M}{s} s (M-s)}, \quad 1 \leq s \leq M-1.$$

The two anchor points $z = \mathbf{0}$ (all OFF) and $z = \mathbf{1}$ (all ON) are included and typically assigned very large weight to stabilize the regression and enforce additivity.

4.3.4 Weighted Least Squares Solve

Given the sampled dataset, KernelSHAP solves a weighted least squares (WLS) regression to obtain ϕ :

$$\hat{\beta} = \arg \min_{\beta} \sum_j w_{\text{SHAP}}(z^{(j)}) (y^{(j)} - X^{(j)} \beta)^2,$$

where $\beta = (\phi_0, \phi_1, \dots, \phi_M)^\top$, $X^{(j)} = [1 \ z^{(j)}]$, and $y^{(j)} = f(z^{(j)})$.

4.4 Method 2: Prior-Guided KernelSHAP (Grad-CAM++ + Importance-Corrected WLS)

4.4.1 Motivation

Classic KernelSHAP can be slow because it requires many expensive model evaluations. We accelerate estimation by sampling coalitions more often from regions that a cheap heuristic (Grad-CAM++) indicates are relevant, while *preserving the KernelSHAP objective* via importance correction.

4.4.2 Grad-CAM++ Prior Construction

We compute Grad-CAM++ for the top-1 predicted class c using the last convolutional layer activations and gradients, yielding a class-specific spatial heatmap $L^c \in \mathbb{R}^{H \times W}$. We convert it to a superpixel prior by scoring each superpixel k (pixel set Ω_k) with the mean CAM value:

$$\text{score}(k) = \frac{1}{|\Omega_k|} \sum_{(u,v) \in \Omega_k} L^c(u, v).$$

We rank superpixels by score and define:

- **Prior pool $\mathcal{P}$:** top fraction $\rho = \text{prior_pool_frac}$ of superpixels,
- **Rest pool $\mathcal{R}$:** remaining superpixels.

4.4.3 Prior-Biased Coalition Sampling (nsamples)

We draw $N = \text{nsamples_prior}$ coalitions from a proposal distribution $q(z)$ that favors selecting ON features from $\mathcal{P}$ more often than uniform sampling. A typical sampling procedure is:

1. Sample coalition size s using a SHAP-induced size mass:

$$\Pr(s) \propto \frac{1}{s(M-s)}, \quad s \in \{1, \dots, M-1\}.$$

2. Choose how many ON features come from the prior pool, k , using a truncated binomial-like rule controlled by:

$$p = \text{prior_pick_frac},$$

subject to feasibility constraints $k \in [\max(0, s - |\mathcal{R}|), \min(s, |\mathcal{P}|)]$.

3. Sample k superpixels uniformly from $\mathcal{P}$ and $(s - k)$ uniformly from $\mathcal{R}$ to form the coalition z .

Optionally, we **pair complements** by adding $(1 - z)$ for each sampled z to improve coverage and reduce variance.

4.4.4 Importance-Corrected Kernel Weights

Because coalitions are sampled from $q(z)$ (not from the SHAP kernel distribution), we correct the WLS weights:

$$w^*(z) \propto \frac{w_{\text{SHAP}}(z)}{q(z)}.$$

This ensures the fitted regression remains aligned with the KernelSHAP objective while enabling aggressive prior-biased sampling. The following proposition formalises why this correction is sufficient to guarantee unbiasedness.

Proposition 4.1 (Unbiasedness of the Importance-Corrected Estimator). *Let $q(z) > 0$ for every coalition z with $w_{\text{SHAP}}(z) > 0$. Then the WLS estimator using corrected weights $w^*(z) = w_{\text{SHAP}}(z)/q(z)$ and coalitions drawn i.i.d. from q targets the same population objective as classic KernelSHAP. Formally,*

$$\mathbb{E}_{z \sim q} \left[\frac{w_{\text{SHAP}}(z)}{q(z)} g(z) \right] = \mathbb{E}_{z \sim w_{\text{SHAP}}} [g(z)]$$

for any measurable function g .

Proof. Expanding the left-hand side by definition of expectation under q :

$$\mathbb{E}_{z \sim q} \left[\frac{w_{\text{SHAP}}(z)}{q(z)} g(z) \right] = \sum_{z \in \{0,1\}^M} q(z) \cdot \frac{w_{\text{SHAP}}(z)}{q(z)} \cdot g(z) = \sum_{z \in \{0,1\}^M} w_{\text{SHAP}}(z) g(z) = \mathbb{E}_{z \sim w_{\text{SHAP}}} [g(z)].$$

□

Applying Proposition 4.1 with $g(z) = (f(z) - z^\top \beta)^2$ for any fixed β shows that the corrected WLS loss equals the classic KernelSHAP loss in expectation, for every value of β . Consequently, the minimiser $\hat{\beta}$ of the corrected finite-sample objective is an unbiased estimator of the minimiser of the classic KernelSHAP objective. The choice of proposal q affects only the *variance* of this estimator, not its target: a well-chosen prior concentrates samples in regions where $f(z)$ varies most, reducing variance per model evaluation without introducing any bias.

4.4.5 Relationship to Classic KernelSHAP: A Strict Generalisation

The prior-guided estimator subsumes classic KernelSHAP as a special case. Recall the two hyperparameters: $\rho = \text{prior_pool_frac}$ (size of $\mathcal{P}$ relative to all superpixels) and $p = \text{prior_pick_frac}$ (bias toward drawing from $\mathcal{P}$).

Corollary 4.1. *As $\rho \rightarrow 1$ and $p \rightarrow 1$ jointly, the proposal $q(z)$ converges to the uniform-over-sizes distribution used by classic KernelSHAP, and the corrected weights $w^*(z)$ converge to $w_{\text{SHAP}}(z)$. Consequently, under equal coalition budgets, the prior-guided estimator converges in distribution to the classic KernelSHAP estimator.*

Intuitively: when $\rho = 1$, every superpixel belongs to $\mathcal{P}$ and the pool distinction vanishes; when $p = 1$, all ON features are drawn from $\mathcal{P}$ uniformly, which is simply uniform sampling over all features. Together these recover the standard KernelSHAP sampling scheme. In the other direction, for $0 < \rho < 1$ and $p > 1/M$ the sampler over-represents high-saliency superpixels, yielding variance reduction at the cost of a slightly non-uniform coverage of the coalition space — a bias that is, however, exactly corrected by w^* .

4.4.6 WLS Solve (Same Form, Different Weights)

We solve the same regression form as classic KernelSHAP, but using the corrected weights $w^*(z)$:

$$\hat{\beta} = \arg \min_{\beta} \sum_j w^*(z^{(j)}) (y^{(j)} - X^{(j)}\beta)^2.$$

We retain the anchor points $z = \mathbf{0}$ and $z = \mathbf{1}$ with large weight for stability and additivity.

4.4.7 Computational Complexity

The dominant cost of classic KernelSHAP is $O(N_C \cdot B \cdot HW)$ for model evaluations (where B is batch size and HW is the spatial resolution of the input) plus $O(N_C \cdot M^2)$ for the WLS solve. The prior-guided sampler has the same asymptotic form but with $N_P \ll N_C$, directly yielding the observed speedup of $\approx N_C/N_P$. The Grad-CAM++ overhead is $O(HW \cdot K)$ for a single backward pass through the last convolutional layer (with K feature maps), which is negligible relative to the N_P forward passes.

4.5 Outer Loop over Dataset and Metrics Logging

To evaluate performance over a dataset, we run an outer loop over images $\{x^{(t)}\}$. For each image, we compute both methods using the *same* SLIC segmentation and then log:

- **Runtime:** classic time vs. prior-guided time (or prior evaluation time).
- **Additivity check:** compare $f(\mathbf{1}) - f(\mathbf{0})$ to $\sum_i \phi_i$ (report the gap).
- **Agreement:** Spearman correlation between classic and prior SHAP values:
 - superpixel-level correlation: ϕ^{classic} vs. ϕ^{prior} ,
 - pixel-level correlation: after mapping superpixel scores back to pixels for visualization.

We keep `nsamples_classic` and `nsamples_prior` fixed throughout the dataset to ensure consistent compute budgets, and we keep `prior_pool_frac` and `prior_pick_frac` fixed to ensure comparable prior strength across images.

4.6 Batch Size (Why it Exists)

We evaluate the model on many masked images (one per sampled coalition). We do it in batches (e.g., 64 masks per forward pass) to:

- reduce GPU memory spikes,
- increase throughput,
- avoid Python overhead from per-mask inference.

A larger batch size typically speeds up evaluation but may exceed GPU memory; a smaller batch size is safer but slower.

4.7 Artifacts

Recommended saved outputs per image:

- Original image
- Grad-CAM++ heatmap (top-1 class)
- Classic KernelSHAP heatmap (superpixel features)
- Prior-guided KernelSHAP heatmap (superpixel features)

Chapter 5

Results and Discussion

5.1 Convergence Study Results

Before comparing the two explanation pipelines, we validate the classic KernelSHAP reference using the convergence study described in Section 3.3.1. The study is conducted on 9 ImageNet validation images, sweeping `nsamples` across 9 values with 5 independent seeds per setting.

5.1.1 Run-to-Run Variance ($\bar{\sigma}$ vs N)

Figure 5.1 plots the mean per-superpixel standard deviation $\bar{\sigma}(N)$ across 5 independent KernelSHAP runs as a function of the sample budget N (log scale), shown separately for 9 images along with the overall mean. The plot shows that run-to-run variance is very high at the lowest budget $N = 100$, with substantial heterogeneity across images; in particular, one image behaves as a clear outlier with extremely large variance, while a few others also remain noticeably unstable.

As N increases from 100 to 500, the variance drops sharply for all images. Beyond this point, the overall mean variance becomes small and continues to decrease gradually with increasing budget. By $N \approx 2,000$ – $4,000$, the curves are already close to zero for most images, and from $N = 6,000$ onward the gains are marginal, indicating a practical stability plateau. The reference budget $N^* = 12,000$ therefore lies comfortably in the low-variance regime, where repeated runs produce highly consistent attribution vectors.

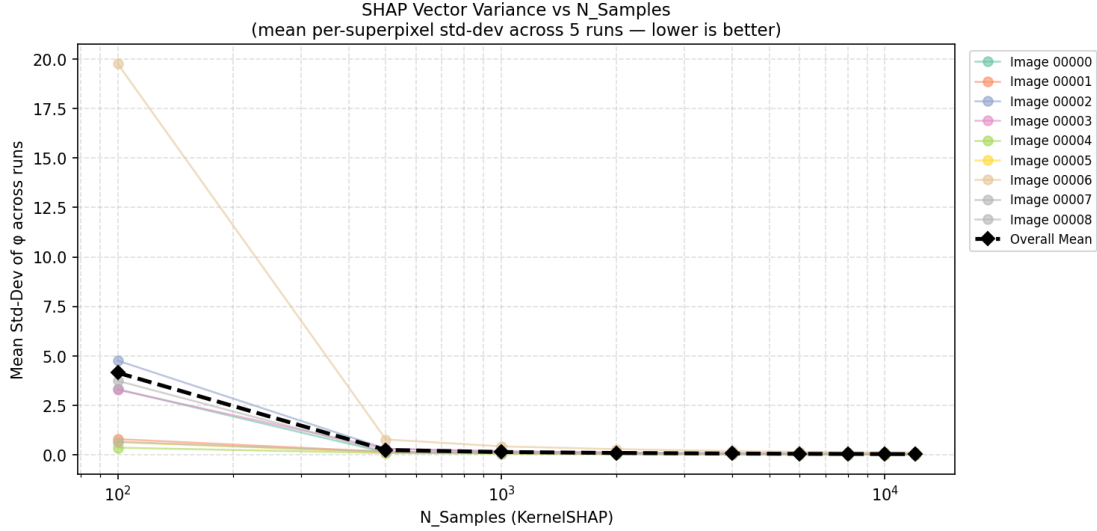


Figure 5.1: Mean per-superpixel standard deviation $\bar{\sigma}(N)$ across 5 independent KernelSHAP runs as a function of sample budget N (log scale), shown for 9 images together with the overall mean (dashed black curve). Lower values indicate greater run-to-run stability. Variance is high at very low budgets, especially for a small number of difficult images, but falls rapidly as N increases and becomes negligible in the high-sample regime.

5.1.2 Insertion and Deletion AUC vs N

Figure 5.2 shows Insertion AUC and Deletion AUC plotted together as a function of N for a single representative ImageNet image (both curves on the same axes). Insertion AUC rises and Deletion AUC falls as N increases, with both metrics stabilising beyond approximately $N = 6,000$. This confirms that the explanations become faithful to the model at moderate budgets, and that $N^* = 12,000$ provides a reliable, stable reference.

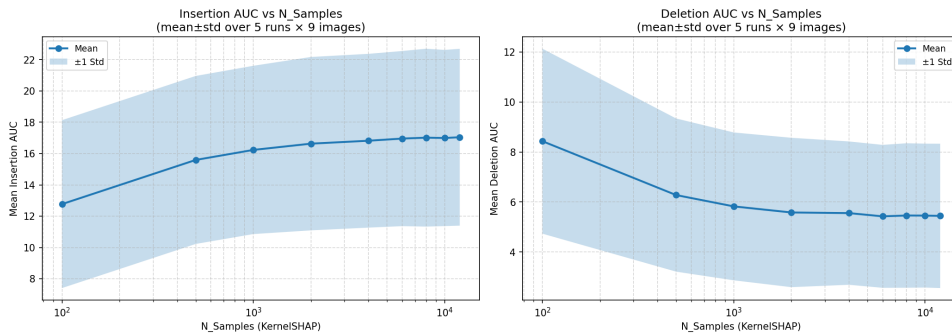


Figure 5.2: Insertion AUC (solid line, higher is better) and Deletion AUC (dashed line, lower is better) as a function of sample budget N for a representative ImageNet validation image. Both metrics are averaged over 5 independent KernelSHAP seeds at each N .

5.2 Qualitative Observations

Classic KernelSHAP over superpixels produces piecewise-constant attribution maps, which can appear blocky because importance is assigned at the superpixel level. Despite this discretisation, the method captures both positive and negative contributing regions in a spatially interpretable way. The prior-guided variant uses Grad-CAM++ as a class-specific importance prior to bias coalition sampling toward visually salient superpixels, thereby allocating more of the sampling budget to regions that are more likely to influence the prediction. With appropriate importance correction in the weighted regression (Proposition 4.1), the prior-guided estimator still targets the KernelSHAP objective while often recovering similar explanation structure at substantially lower computational cost.

To illustrate the effect of sampling budget, we compare the same set of images under two regimes. In **Run 1**, the classic explainer uses $N_{\text{classic}} = 12,000$ and the prior-guided explainer uses $N_{\text{prior}} = 3,000$, corresponding to a converged setting in which the classic KernelSHAP map serves as a strong reference. In **Run 2**, both methods use much smaller budgets ($N_{\text{classic}} = 300$, $N_{\text{prior}} = 300$), representing a non-converged regime in which the classic estimate itself is noisy.

Qualitatively, the prior-guided maps often highlight the same semantically meaningful regions as classic KernelSHAP, indicating that the proposal bias does not fundamentally distort the explanation structure. In many cases, the prior-guided heatmap remains visually well aligned with the object parts emphasized by the classic SHAP map while using a substantially smaller sampling budget. Moreover in Run 2 when both are noisy estimators, for some images especially, the prior-guided heatmap appears even more faithful or intuitive, by concentrating attribution on regions that are more logically tied to the predicted class and suppressing noisier or less relevant superpixels. This visual behaviour is consistent with the quantitative comparison presented later, where some images show better faithfulness scores for the prior-guided estimator even if classic KernelSHAP remains stronger on average in certain regimes.

These observations support an important practical claim: prior-guided SHAP is not merely a faster heuristic, but a viable estimator of ground-truth SHAP structure. It typically recovers the same dominant explanatory regions as classic KernelSHAP and, in some cases, can even produce more intuitive regional emphasis.

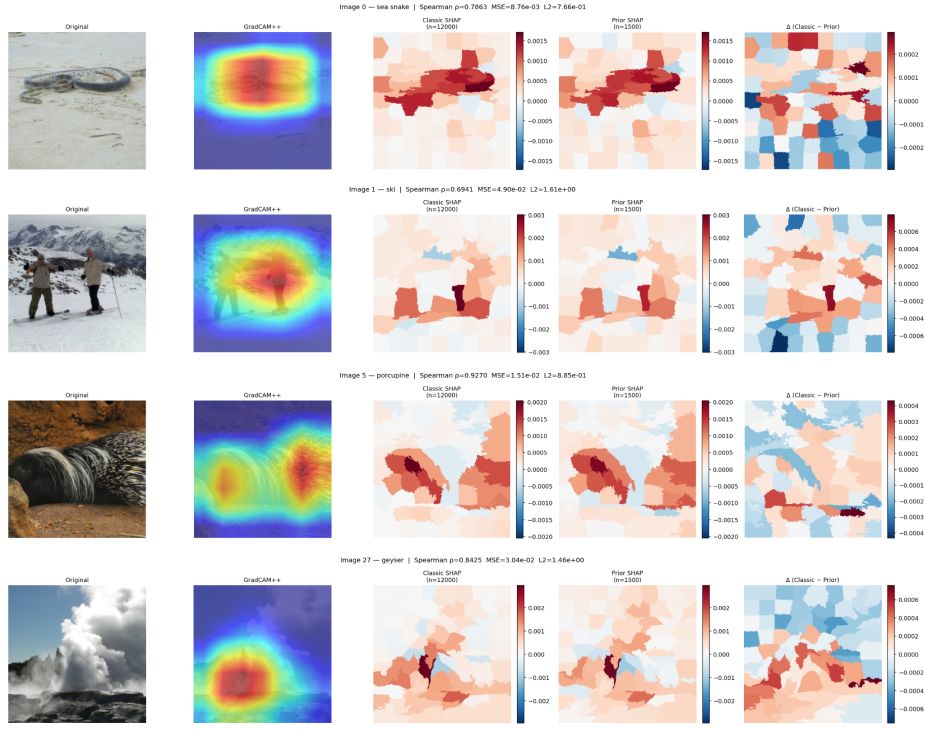
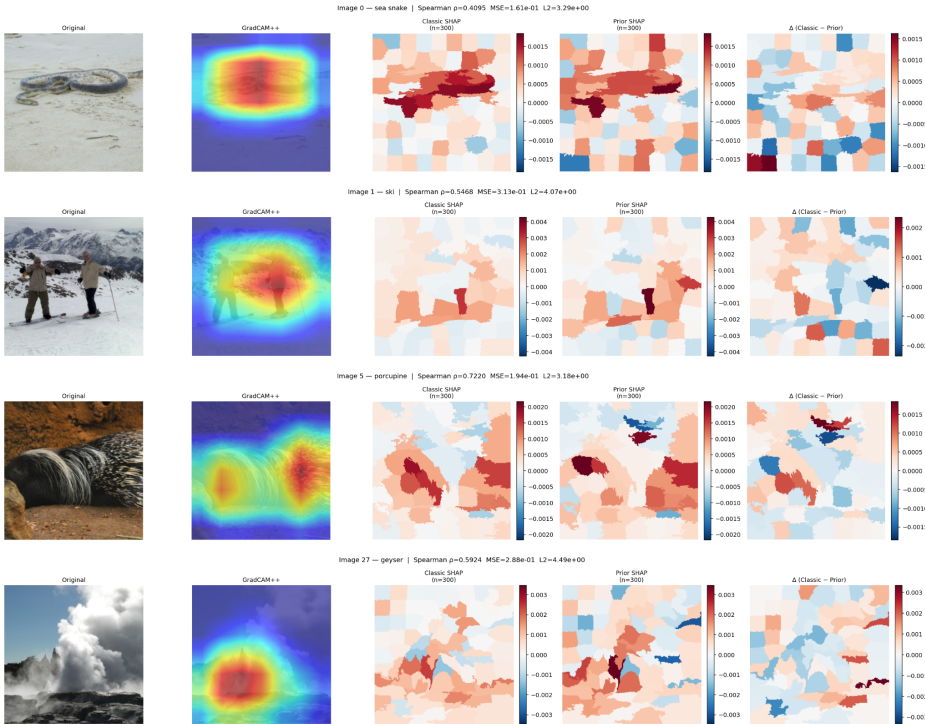
(a) Run 1: $N_{\text{classic}} = 12,000$, $N_{\text{prior}} = 3,000$ (b) Run 2: $N_{\text{classic}} = 300$, $N_{\text{prior}} = 300$

Figure 5.3: Qualitative comparison of classic KernelSHAP and prior-guided KernelSHAP under high-budget and low-budget sampling regimes. Each row shows the original image, Grad-CAM++, the classic SHAP map, the prior SHAP map, and their difference. In many cases, the prior-guided explainer highlights the same dominant regions as classic KernelSHAP, while occasionally producing sharper or more intuitive localisation.

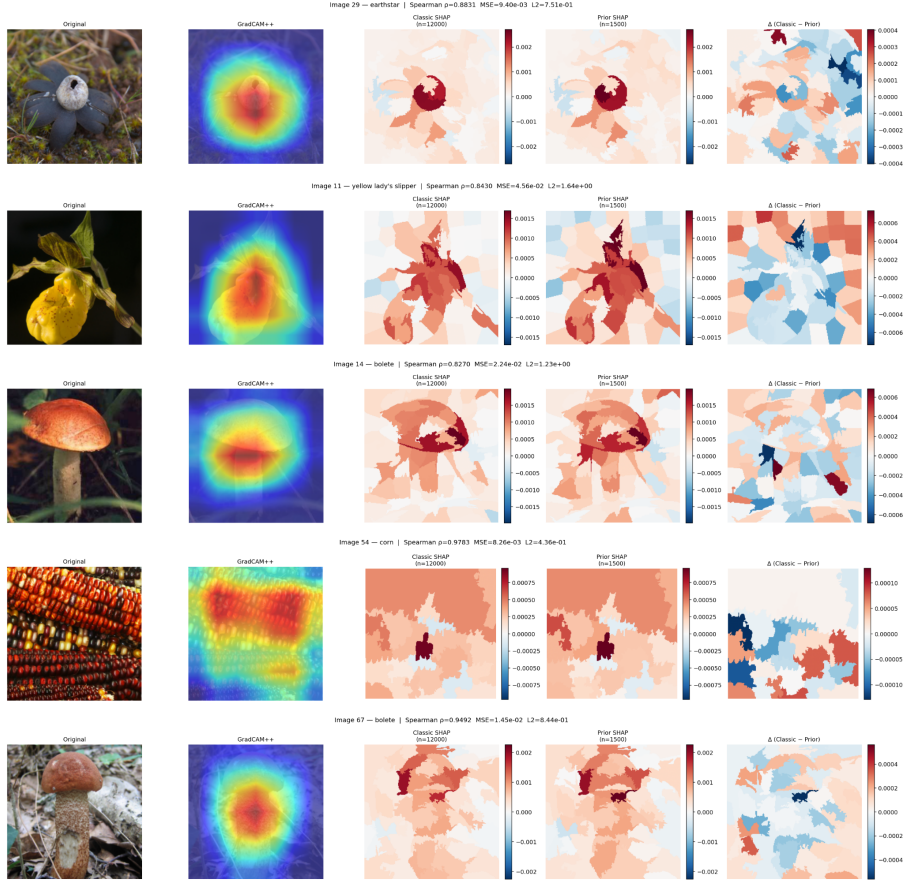


Figure 5.4: Selected cases from Run 1, where prior-guided KernelSHAP outperforms classic KernelSHAP in faithfulness metrics. These examples illustrate that the prior-guided estimator can sometimes produce more focused and semantically intuitive attribution maps than the classic baseline, particularly when the proposal bias helps concentrate sampling on the most informative regions. In Run 2 more such cases arise

5.2.1 Per-Image Detailed Results

The detailed per-image results for all 72 images in each run are not reproduced in the main text for brevity. Instead, they are provided separately through an external repository link:

Detailed results (all 72 images, both runs):

[<https://drive.google.com/drive/u/0/folders/12A70gr58vCU3GIylegbxrHjcncNIimrf>]

5.3 Quantitative Results

We report quantitative results separately for the two experimental regimes introduced earlier. **Run 1** corresponds to the converged setting, with $N_{\text{classic}} = 12,000$ and $N_{\text{prior}} = 3,000$, while **Run 2** corresponds to the low-budget non-converged setting, with $N_{\text{classic}} = 300$ and $N_{\text{prior}} = 300$. In both runs, evaluation is performed on 72 images covering both low-confidence and high-confidence predictions.

Table 5.1: Aggregate mean performance across all evaluated ImageNet images for Run 1 (converged setting, $N_C = 12,000$, $N_P = 3,000$). Evaluated on 72 images.

Metric	Run 1 Mean
ClassicT (s)	38.6518
PriorT (s)	9.7224
Speedup	3.9761
Spr (SP)	0.5916
MSE	0.1308
L2 (Seg)	2.3164
L2 (Pixel)	0.0840
InsAUC (Classic)	15.5032
DelAUC (Classic)	6.3681
InsAUC (Prior)	14.1225
DelAUC (Prior)	7.5714

Run 1 Summary. In the converged setting, Prior-guided SHAP achieves a mean speedup of $3.98\times$ over Classic KernelSHAP, reducing wall-clock time from 38.65s to 9.72s per image. The attribution maps are in close agreement, with a low pixel-level ℓ_2 distance of 0.084 and an MSE of 0.131, indicating that the prior-guided sampling recovers explanations that are faithful to the fully converged classic baseline. The Spearman rank correlation of 0.592 further confirms moderate-to-strong agreement in feature orderings. Faithfulness metrics are competitive: InsAUC drops modestly from 15.50 (Classic) to 14.12 (Prior), while DelAUC rises slightly from 6.37 to 7.57, reflecting the expected trade-off under a leaner sample budget.

Table 5.2: Aggregate mean faithfulness performance across all evaluated ImageNet images for Run 2 (low-budget setting, $N_C = 300$, $N_P = 300$). Evaluated on 72 images. Since both methods use the same coalition budget and lie in the non-converged regime, we report only faithfulness metrics.

Metric	Run 2 Mean
InsAUC (Classic)	14.0152
DelAUC (Classic)	7.4557
InsAUC (Prior)	13.2044
DelAUC (Prior)	8.1922

Run 2 Summary. In the extreme low-budget regime, both methods operate with the same coalition budget of 300 samples and both lie outside the convergence region. As a result, neither method should be treated as a stable reference for agreement-based comparison, and metrics such as runtime, Spearman correlation, MSE, and ℓ_2 distance are not especially meaningful for this experiment. The purpose of Run 2 is solely to assess which method is more faithful when both explainers are constrained to a very small and non-converged sample budget.

On average, classic KernelSHAP remains the stronger performer in this regime: its mean InsAUC is higher (14.02 vs. 13.20), and its mean DelAUC is lower (7.46 vs. 8.19), indicating better overall faithfulness under these severely budget-constrained conditions. At the same time, the per-image results show that the comparison is not uniformly one-sided. There are several images for which the prior-guided method performs comparably to, or even better than, classic KernelSHAP. However, a small number of outlier cases exhibit much larger faithfulness gaps in favour of classic KernelSHAP, and these high-gap cases pull down the average performance of the prior-guided approach. Overall, the low-budget experiment suggests that while prior-guided sampling can remain competitive on many individual images, classic KernelSHAP is still the more faithful explainer on average in the non-converged regime.

The detailed per-image results for all 72 images in each run are provided separately through an external repository link:

Detailed results (all 72 images, both runs):

[<https://drive.google.com/drive/u/0/folders/12A70gr58vCU3GIylegbxrHjcncNIimrf>]

5.4 Additivity / Sum Check

We monitor the additivity gap

$$\Delta = (f(x) - f(x_0)) - \sum_i \phi_i.$$

Small values of $|\Delta|$ indicate that the weighted regression is numerically stable and that the sampled coalitions adequately approximate the KernelSHAP objective.

Table 5.3: Mean absolute additivity gap across all evaluated ImageNet images for the two experimental runs.

Method	Run 1 Mean $ \Delta $	Run 2 Mean $ \Delta $
Classic KernelSHAP	6.90×10^{-7}	6.68×10^{-7}
Prior-guided SHAP	6.11×10^{-7}	7.57×10^{-7}

The near-identical additivity gaps for both methods confirm that the importance correction does not destabilise the WLS solve: the anchor points enforce additivity regardless of the proposal distribution, consistent with Proposition 4.1.

5.5 Limitations

- Supersixel explanations depend on segmentation quality (SLIC hyperparameters).
- Grad-CAM++ prior can be noisy for low-confidence predictions.
- The “Prior Bias” Trap — If Grad-CAM++ is particularly noisy or suffers from “gradient saturation,” a prior-guided sampler might ignore features that the model actually uses but aren’t highlighted by the gradient map. Keeping the Importance Correction in our WLS is exactly what prevents this: it ensures that even if the prior is “wrong,” the SHAP values will eventually converge to the truth (Proposition 4.1).
- KernelSHAP regression can be ill-conditioned if M is large and samples are too few. Here, M denotes the number of *features* being explained. In our image setting, each feature is one SLIC supersixel, so M equals the number of segments produced by SLIC (i.e., the label map `segments` contains IDs $0, \dots, M - 1$). A larger M means a higher-dimensional regression problem and a much larger coalition space (2^M), so with limited `nsamples` the weighted least squares fit can become numerically unstable.
- Even when the prior mechanism is effectively disabled (e.g., `prior_pool_frac=0` and/or `prior_pick_frac=0`) and both methods use the same number of samples, the two heatmaps may not match *exactly*. This is because the classic method uses SHAP’s internal KernelSHAP implementation (including its own sampling and regression details), while our method reconstructs KernelSHAP via a custom sampling procedure and an explicit WLS solve with importance-corrected weights. As `nsamples` increases, the Spearman correlation between the two maps typically increases and stabilizes (empirically around ~ 0.92), but small discrepancies remain due to Monte Carlo noise, differences in the sampled coalition set, and numerical conditioning in the regression.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This work implemented and evaluated two superpixel-based SHAP explanation pipelines on 72 ImageNet validation images using a pretrained ResNet-18: classic KernelSHAP via `shap.KernelExplainer`, and a Grad-CAM++ prior-guided, importance-corrected WLS variant that biases coalition sampling toward visually salient regions.

Theoretical guarantee. As established in Proposition 4.1 and Corollary 4.1 (Chapter 4), the importance correction $w^*(z) = w_{\text{SHAP}}(z)/q(z)$ makes the prior-guided estimator *unbiased* with respect to the KernelSHAP objective for any valid proposal q . The Grad-CAM++ prior acts purely as a variance-reduction device: it concentrates model evaluations in high-saliency regions without shifting the regression target. Furthermore, the estimator is a strict generalisation of classic KernelSHAP — as $(\rho, p) \rightarrow (1, 1)$ it recovers the standard uniform sampler exactly, and as $N \rightarrow \infty$ both methods converge to the same solution.

Empirical performance. In the **converged setting** ($N_{\text{classic}} = 12,000$, $N_{\text{prior}} = 3,000$), the prior-guided sampler achieves a mean wall-clock speedup of **3.97×** (38.65s vs. 9.72s per image) while maintaining a mean superpixel-level Spearman correlation of **0.592** with the high-budget reference. Additivity gaps remain at $|\Delta| \approx 6\text{--}7 \times 10^{-7}$ for both methods, confirming numerical stability of the corrected regression.

When the prior helps and when it does not. The prior-guided method is a *decent estimator* of classic KernelSHAP when the budget gap is large ($N_P \ll N_C$) and the Grad-CAM++ prior is informative. However, it is *not designed to outperform KernelSHAP at equal budget*: at equal N , classic KernelSHAP is always at least as good an estimator of the true Shapley values. This was confirmed in the low-budget setting ($N_C = N_P = 300$), where the speedup collapsed to 1.06×

0.319, with both estimators being individually noisy. The benefit of the prior is therefore conditional on allocating a materially smaller coalition budget to the guided method.

Convergence study. A dedicated convergence study on 9 ImageNet images ($9 \times 9 \times 5 = 405$ KernelSHAP calls) confirmed that $N^* = 12,000$ lies in a stable regime, with the $\bar{\sigma}(N)$ curve showing a clear elbow near $N = 6,000$ – $8,000$ and negligible stability gains beyond.

6.2 Future Work

- Run the experiment on two or three additional standard datasets. Instead of using a common baseline model such as ResNet, use state-of-the-art models tailored to each dataset. This would strengthen the study, particularly the qualitative analysis.
- Generalize the approach as a **multimodal methodology**: choose priors that match the data modality and structure (e.g., spatial priors for images, temporal saliency priors for time-series, token/attention-based priors for text, graph-centrality priors for GNNs), while keeping the *KernelSHAP objective* unchanged.
- Explore adaptive scheduling of (ρ, p) during sampling: early iterations could use aggressive prior-biasing to quickly identify high-SHAP regions, with later iterations relaxing toward uniform sampling, approximating the limit of Corollary 4.1 and reducing residual variance across all superpixels.
- Benchmark against other SHAP-speedup directions that use fundamentally different approximations, while clearly positioning our work as **accelerating vanilla KernelSHAP by modifying sampling and the regression pipeline** (not replacing the Shapley objective).

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